

Enterprise Systems Integration Policy

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APPROVED — ACTIVE

1. Purpose & Scope

This policy establishes the standards, architecture requirements, and governance processes for connecting business systems, APIs, endpoints, and third-party services to the TechCorp Enterprise AI Platform. It ensures that every integration is secure, maintainable, observable, and aligned with the enterprise data flow architecture managed by IT in coordination with the **Engineering Department**.

This policy applies to all internal system-to-system integrations, external third-party service connections, API exposures, webhook configurations, and data pipelines that interact with the Enterprise AI Platform. It applies to IT, Engineering, and any department procuring or building integrations that touch the platform.

2. AI Platform Integration

The IT Department governs enterprise integrations through an API gateway and integration middleware layer that is itself monitored by the Enterprise AI Platform's observability suite, with security enforcement from the **Security Department**. Core integration components include:

- API Gateway & Catalog** — Centralized API management layer where all platform-facing endpoints are registered, versioned, rate-limited, and monitored. The AI Platform's integration discovery engine automatically catalogs new endpoints and flags unregistered APIs for IT review.
- Integration Health Monitor** — AI-driven observability across all active integrations, tracking latency, error rates, data volume anomalies, and schema drift, with automated alerts to the owning team and IT when an integration degrades below defined thresholds.
- Third-Party Connector Registry** — Vetted library of pre-approved connectors for common enterprise services (CRM, ERP, HRIS, data warehouses), each with documented data flow scope, security posture, and compliance certifications validated jointly with **Security** and **Legal**.

- **Data Flow Governance Engine** — AI-assisted mapping of all data flows between integrated systems, enforcing data classification boundaries and flagging unauthorized cross-tier data movement for review by the owning department and IT.

Connected Departments: Engineering (API development & service ownership) • Security (integration security review & data flow enforcement) • Legal (third-party data processing agreement compliance) • AI (integration monitoring model accuracy) • Operations (business process integration health)

3. Permitted Integration Activities

Activity	AI Role	Human Oversight Required
API registration & publishing	Auto-discovery & catalog population	Service owner confirms API metadata and deprecation timeline
Third-party connector onboarding	Pre-vetting risk scoring & compliance mapping	IT, Security, and Legal jointly approve before production activation
Integration health monitoring	Anomaly detection & degradation alerting	Integration owner triages alerts; IT escalates unresolved degradations
Data flow mapping & boundary enforcement	Automated lineage tracking & cross-tier flagging	Data steward reviews quarterly; Legal reviews any cross-border flows
API version deprecation & sunseting	Consumer impact analysis & migration scheduling	Service owner approves deprecation timeline; IT validates consumer migration
Integration performance optimization	Bottleneck detection & caching recommendation	Engineering reviews and implements recommendations

4. Prohibited Activities

The following are strictly prohibited under this policy:

- **Unregistered production integrations.** Every integration touching the Enterprise AI Platform must be registered in the API Gateway and Catalog. Unregistered API endpoints or data pipelines in production are prohibited and subject to immediate quarantine by IT.
- **Direct database access bypassing the API layer.** External systems and services must integrate through approved APIs or managed connectors. Direct database connections from external systems, including read-only access, are prohibited without a documented exception approved by IT and Security.

- **Hard-coded credentials or secrets in integration code.** All integration authentication must use the centralized secrets manager and identity provider. Embedded credentials are prohibited and will be flagged by the AI code scanner during the integration review process.
- **Cross-tier data flow violations.** Integrations must not move data from a higher classification tier to a lower tier without explicit approval from the data steward and Legal. The Data Flow Governance Engine enforces this automatically.
- **Unauthorized third-party data egress.** Integrations sending TechCorp data to external services must have an approved Data Processing Agreement on file with Legal, verified by the Third-Party Connector Registry before activation.

5. Integration Architecture & Data Governance

All integrations must comply with TechCorp's **AI Safety Guidelines** (AI Department) and the **Data Protection Standards** (Security Department). Architecture requirements include:

- All integrations must use the centralized API Gateway for authentication, rate limiting, and request logging. Service-to-service authentication must use certificate-based mutual TLS or managed identity tokens issued by the identity provider.
- Every integration must define a data flow scope document specifying source system, target system, data classification tier, data volume expectations, and retention requirements before deployment.
- All API endpoints must be versioned following semantic versioning; breaking changes require a new major version with the previous version supported for a minimum 90-day deprecation window communicated through the API Catalog.
- Third-party integrations must include a documented failover plan: what happens to dependent business processes if the external service is unavailable, with degraded-mode operation defined and tested annually by the owning department and IT.
- Integration logs must flow into the centralized observability platform with a minimum 90-day retention for operational data and 12-month retention for audit-relevant data, per the retention schedule maintained by Legal.

6. Integration Quality Standards

Personnel building or managing integrations must adhere to TechCorp's **Prompt Engineering Standards** when using AI-assisted integration tooling. Integration-specific requirements include:

- API documentation prompts must include the endpoint purpose, authentication method, request/response schema, rate limits, and error codes for the AI to

generate accurate and complete catalog entries.

- Integration health monitoring prompts must be tuned per integration with expected latency baselines, data volume profiles, and business criticality context for accurate anomaly detection.
- Third-party connector evaluation prompts must reference the vendor’s security certifications, data processing terms, and integration scope to produce calibrated risk scores for IT and Legal review.
- All integration code and configuration must be version-controlled in the approved repository; AI-generated integration configurations must be peer-reviewed before deployment to production.

7. Roles & Responsibilities

Role	Responsibility
VP of IT	Overall integration governance ownership; API Gateway infrastructure budget; approval of enterprise-wide integration architecture changes.
Integration Architect	Integration pattern standards; API design review; connector library curation; cross-system data flow architecture; integration review board facilitation.
Service Owners (Engineering)	API development and versioning; endpoint registration in API Catalog; integration performance monitoring; deprecation and migration planning.
Security Department	Integration security review; data flow boundary enforcement; third-party security assessment; secrets management governance; integration penetration testing.
Legal Department	Third-party Data Processing Agreement review; cross-border data flow approval; regulatory compliance for integrated data; API terms of use governance.
AI Department	Integration monitoring model accuracy; API discovery engine maintenance; data flow anomaly detection tuning; integration AI tool prompt standards.
Department Integration Sponsors	Business justification for new integrations; data flow scope definition; integration failover plan ownership; annual integration health recertification.

8. Incident Response

Any of the following constitutes an integration incident and must be reported within the timeframe specified:

- Integration causing data corruption or loss in a downstream system — report within 30 minutes to IT, the affected system owner, and the data steward; halt the integration immediately.

- Unauthorized data egress through an integration — Security receives real-time AI alert; IT blocks the integration endpoint within 15 minutes; Legal determines notification obligations.
- API Gateway outage affecting multiple integrations — IT detects within 2 minutes; all department leads notified within 15 minutes; break-glass direct-service routing activated per the continuity runbook.
- Cross-tier data flow violation detected by the Data Flow Governance Engine — report within 1 hour to the data steward, IT, and Security; integration quarantined until the data flow is corrected and approved.
- Third-party connector security vulnerability disclosure — Security assesses severity within 4 hours; IT disables affected connector if critical; owning department activates failover plan.

All integration incidents are logged in the platform's incident management module. Incidents involving data exposure or third-party connector compromise require a post-incident review within 5 business days involving IT, Security, Legal, and the owning department. Material data egress incidents are escalated to the General Counsel within 24 hours.

9. Training & Compliance

- All Engineering personnel building platform integrations must complete the API Design & Integration Standards workshop before deploying their first production integration.
- IT personnel managing the API Gateway and integration infrastructure must complete the Enterprise Integration Governance certification within 60 days of onboarding.
- Department sponsors proposing new integrations must complete the Integration Request & Data Flow Scoping training covering the approval workflow, data classification rules, and failover planning requirements.
- Quarterly integration health reviews are conducted by IT with each department; unregistered or non-compliant integrations are flagged for remediation or decommissioning.
- Non-compliance with this policy, including deploying unregistered integrations, is subject to integration quarantine and progressive discipline administered by HR.

10. Review & Revision

This policy is reviewed every 6 months by the VP of IT, Integration Architect, and Security Department representative, with input from the Engineering Department (for API development practices and service ownership), Legal Department (for third-party data processing and cross-border flow requirements), AI Department (for integration

monitoring tooling and AI discovery accuracy), and Operations Department (for business process integration health). Revisions are versioned, approved by the VP of IT, and synchronized with updates to the Enterprise AI Platform Operations Policy, the Data Protection Standards, and the IT Access Management Policy. Emergency revisions triggered by integration-related incidents or significant API Gateway architecture changes follow an accelerated 14-day review cycle.

Referenced Policies: Enterprise AI Platform Operations Policy (IT-OPS-001) • Identity, Access & Provisioning Policy (IT-IAM-001) • Data Protection Standards (Security Department) • AI Safety Guidelines (AI Department) • Prompt Engineering Standards (AI Department) • Corporate Compliance & Data Governance Policy (LEG-CORP-002)